

# MAHADEV CHAVAN

## Data Scientist / AI Engineer

+91 8149287361   [mahadev.c2019@gmail.com](mailto:mahadev.c2019@gmail.com)   [linkedin.com/in/mahadev-chavan](https://linkedin.com/in/mahadev-chavan)   [github.com/mahadevchavan](https://github.com/mahadevchavan)

Data Science Engineer with 3+ years of experience designing and deploying production-grade AI systems across GenAI, NLP, Computer Vision, and Predictive Analytics. Experienced in building RAG pipelines, agentic AI workflows, scalable ML APIs, and cloud-native deployments using AWS, Docker, and Kubernetes.

## Work Experience

**Data Science Engineer – Ksolves India Ltd** Pune, India

2024 – Present

- **RAG-based Estimation Automation:** Architected and deployed a production-grade RAG platform using LangChain, vector search, LLM APIs, and FastAPI, reducing manual effort by 75%.
- **HVAC Predictive Maintenance:** Engineered and deployed a predictive maintenance solution using XGBoost and anomaly detection on HVAC telemetry data, enabling proactive fault identification and achieving approximately 50% earlier failure detection.
- **Salesforce Ticket Routing System:** Architected and deployed an ML-driven ticket assignment system integrated with Salesforce, leveraging historical resolution data and workload balancing to automate routing decisions and reduce manual effort by approximately 70%.
- **AI Proctoring Platform:** Implemented computer vision pipelines using facial landmarks and gaze estimation to automate candidate engagement monitoring during coding assessments.

**Trainee Engineer – Neosoft Technologies Pvt Ltd** Pune, India

2022 - 2023

- Implemented data ingestion and preprocessing workflows for structured and unstructured datasets sourced from databases, APIs, and external systems to support enterprise analytics initiatives.
- Engineered data wrangling and exploratory analysis workflows using Pandas, SQL, and Matplotlib to uncover actionable operational insights.
- Implemented end-to-end data pipelines and computer vision workflows using OpenCV and CNNs, supporting model training, evaluation, optimization, and performance analysis.

**Data Science Intern – Sciffer Analytics Pte Ltd** Pune, India

2021

- Engineered high-quality annotated datasets through large-scale data sourcing, curation, and labeling workflows to support computer vision model development.
- Optimized annotation operations by supervising a 5-member team, ensuring workflow continuity, efficient task allocation, and adherence to data quality standards.

## Projects

**Agentic Knowledge Retrieval System (RAG & Generative AI)**

- Architected an advanced Agentic RAG system utilizing LangGraph to dynamically route complex queries, ingest diverse unstructured data (PDFs, logs, code files), and synthesize context-aware responses.
- Integrated Model Context Protocol (MCP) frameworks to establish secure, standardized communication between LLMs and external systems, improving retrieval efficiency and system modularity.
- Engineered and optimized document parsing, embedding pipelines, and semantic retrieval workflows using vector databases to minimize hallucinations and improve retrieval fidelity.

**Nationwide Distributed Energy Resource Modeling & Analysis**

- Architected and scaled a distributed data pipeline using PySpark and AWS (S3, Glue, Redshift) for nationwide energy analytics, optimizing ingestion, transformation, and reporting workflows.

**Hindi Character Recognition (Deep Learning)**

- Engineered and optimized a CNN-based deep learning model for multi-class Devanagari character recognition, achieving 96% training accuracy and 94% test accuracy.

## Skills & Tools

- **Programming & Core Stack:** Python, SQL, JavaScript, Bash Scripting, Big Query
- **Data Analysis & Visualization:** Pandas, NumPy, Tableau, PySpark, Matplotlib, Power BI
- **Machine Learning:** Supervised Learning, Ensemble Models, Time-Series Forecasting, Feature Engineering, Model Evaluation
- **Deep Learning:** TensorFlow, Keras, CNNs, Computer Vision, NLP
- **GenAI / LLM Systems:** RAG, Agentic AI, LangChain, LangGraph, MCP, OpenAI APIs, Claude, Gemini, Ollama, HuggingFace, FAISS, Pinecone, Langfuse, Guardrails
- **MLOps, Tools & Cloud:** AWS (S3, Glue, Redshift, Lambda), Docker, Kubernetes, Playwright, Git, GCP

- **Others:** FastAPI, Flask, Django, ETL, OOP, Predictive Modeling, Web Scraping, Prompt Engineering

## Education & Learning

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|----------------------------------------------------------------|-------------|------------------|
| <b>PG Diploma in Big Data Analytics – CDAC Pune</b>            | Pune, India | <i>Mar 2024</i>  |
| <b>PG Diploma In AI &amp; ML – University Of Hyderabad</b>     |             | <i>Feb 2022</i>  |
| <b>MSc Computer Science – Savitribai Phule Pune University</b> | Pune, India | <i>June 2020</i> |
| <b>BSc Computer Science – Savitribai Phule Pune University</b> | Pune, India | <i>2018</i>      |
| <b>Certifications:</b>                                         |             |                  |
| AWS Generative AI with LLMs (Coursera, 2025)                   |             |                  |
| AWS ML Scholarship (Udacity, 2020)                             |             |                  |